



Sort Two Ways

Quick facilitation notes & answer key

Goal

Children sort one pile by two different rules — by exact shape, then by round-vs-pointy. Seeing that the same set can be grouped different ways builds flexible classification, a computational-thinking skill (the rule decides the groups).

Kindergarten frame (Ontario)

Problem Solving and Innovating — children classify the same objects by more than one attribute.

No reading required of the child: read each prompt aloud; the child draws into the group boxes.

Materials

The printed worksheet + a pencil/crayon. Optional: a small pile of shape blocks to sort by hand two ways.

Run it (about 15 min)

1. MODEL: name the pile — two circles, two triangles, one star.
2. WAY 1: sort by shape together (Ex 1) — circles, triangles, star.
3. WAY 2: change the rule to round vs pointy (Ex 2) — circles are round; triangles and star are pointy.
4. YOU TRY: children sort the Ex 3 pile both ways. Connect: 'Computers sort by different rules too — by size, colour, or name.'

Answer key (modelled by the run-gate)

Pile ● ▲ ● ★ ▲ — Way 1 (by shape): circle = 2, triangle = 2, star = 1.

Way 2 (round vs pointy): round = 2 (the circles); pointy = 3 (two triangles + one star).

Ex 3 pile ★ ● ▲ ● — by shape: star 1, circle 2, triangle 1. Round/pointy: round 2, pointy 2.

Watch for & differentiate

Common slip: re-sorting Way 2 by shape out of habit — restate the new rule (round vs pointy).

Simplify: do only Way 1 with hands-on blocks, then introduce the second rule.

Extend: invent a third rule together (big vs small, or has-a-point vs no-point).

Movement option: sort the class two ways — by shirt colour, then by shoes on/off.

Success looks like

The child sorts the one pile correctly under both rules and can say that changing the rule changes the groups.